

# Pair Coulomb Density Matrix on a Sphere

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We tabulate the diagonal pair Coulomb density matrix on a sphere.

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## I. INTRODUCTION

We want to find the Pair Coulomb Density Matrix (PCDM) on a Sphere so to extend the work of Ref. [1–3].

We will denote with a Greek index a coordinate basis component and with a Roman index a particle label. We will adopt Einstein summation convention to tacitly assume a sum over repeated Greek indexes. We distinguish between upper contravariant Greek indexes and lower covariant Greek indexes. Passing between the two is accomplished with a contraction with the metric tensor, as usual.

The metric tensor of the Sphere of radius  $a$  is

$$||g_{\alpha\beta}|| = a^2 \begin{pmatrix} 1 & 0 \\ 0 & \cos^2 \theta \end{pmatrix}. \quad (1.1)$$

with  $g = \det||g_{\alpha\beta}|| = a^4 \cos^2 \theta$ .

It is important to realize that on a curved surface the two body problem is generally not separable into a center of mass and a relative one body problem. Therefore, in full generality, we will study the position representation  $\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$  of the density matrix  $\rho = \exp(-Ht)$ , where  $H$  is the Hamiltonian operator for the two bodies. These are initially at  $\mathbf{r}_1$  and  $\mathbf{r}_2$  and at an imaginary time  $t$  later they are at  $\mathbf{r}'_1$  and  $\mathbf{r}'_2$ .

Therefore, the PCDM on the Sphere will satisfy the Bloch equation

$$\frac{\partial \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}{\partial t} = \left( \frac{\lambda_1}{\sqrt{g_1}} \nabla_{1\alpha} \sqrt{g_1} \nabla_1^\alpha + \frac{\lambda_2}{\sqrt{g_2}} \nabla_{2\alpha} \sqrt{g_2} \nabla_2^\alpha - \phi(\mathbf{r}_1, \mathbf{r}_2) \right) \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t), \quad (1.2)$$

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; 0) = \delta(\mathbf{r}_1 - \mathbf{r}'_1) \delta(\mathbf{r}_2 - \mathbf{r}'_2) / \sqrt{g_1 g_2}, \quad (1.3)$$

where, for particle  $i = 1, 2$ ,  $\lambda_i = \hbar^2/2m_i$  with  $m_i$  the mass of particle  $i$ ,  $g_i = a^4 \sin^2 \theta_i$ ,  $\nabla_i = (\nabla_{i1}, \nabla_{i2}) = (\partial/\partial\theta_i, \partial/\partial\varphi_i)$  is the gradient respect to the coordinates  $\mathbf{r}_i = (r_i^1, r_i^2) = (\theta_i, \varphi_i)$ ,  $\delta$  is a two dimensional Dirac delta function,  $\rho$  is the pair density matrix, and  $\phi$  is the Coulomb pair potential.

The Coulomb pair potential for particles moving **on** the surface can be chosen as

$$\phi_{\text{on}}(\mathbf{r}_1, \mathbf{r}_2) = Z_1 Z_2 e^2 / d(\mathbf{r}_1, \mathbf{r}_2), \quad (1.4)$$

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where  $Z_i e$  is the charge of particle  $i$  and for the distance between the two points  $\mathbf{r}_1$  and  $\mathbf{r}_2$ ,  $d(\mathbf{r}_1, \mathbf{r}_2)$ , on the Sphere we can either choose the geodesic distance

$$d_g(\mathbf{r}_1, \mathbf{r}_2) = a \arccos[\sin \theta_1 \sin \theta_2 + \cos \theta_1 \cos \theta_2 \cos(\varphi_1 - \varphi_2)], \quad (1.5)$$

or, viewing the Sphere as embedded in the three dimensional space, the Euclidean distance

$$d_e(\mathbf{r}_1, \mathbf{r}_2) = 2a \sin(d_g/2a). \quad (1.6)$$

For particles living **in** the surface [4] we may choose

$$\phi_{\text{in}}(\mathbf{r}_1, \mathbf{r}_2) = -Z_1 Z_2 e^2 \ln[d_e(\mathbf{r}_1, \mathbf{r}_2)/\ell], \quad (1.7)$$

with  $\ell$  a constant with the dimensions of a distance. We will call  $\phi_{\text{on}}^g$ ,  $\phi_{\text{on}}^e$ ,  $\phi_{\text{in}}$  the three cases just illustrated.

We will measure energy in units of the Hartree  $e^2/a_B$ , with  $e$  the charge of the electron and  $a_B = \hbar^2/m_e e^2$  Bohr radius, with  $m_e$  the mass of the electron. And length in units of the Bohr radius  $a_B$ .

The free particles,  $\phi \rightarrow 0$ , pair density matrix can be determined from the work of Bastianelli and Corradini [5]. We will call their solution for the single particle  $i$ ,  $\rho_{i0}(\mathbf{r}_i, \mathbf{r}'_i; \tau)$ . Therefore the pair free density matrix will be

$$\rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_{10}(\mathbf{r}_1, \mathbf{r}'_1; \tau) \rho_{20}(\mathbf{r}_2, \mathbf{r}'_2; \tau). \quad (1.8)$$

Then our PCDM can be written as

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) = \rho_0(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t) e^{-P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)}, \quad (1.9)$$

where  $P$  is the *inter-action*.

Aim of this work is to tabulate  $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$  for the three Coulomb cases illustrated above,  $P_{\text{on}}^g$ ,  $P_{\text{on}}^e$ ,  $P_{\text{in}}$ . For small  $t \ll 1$ , for each of these cases, the *primitive approximation* [6]

$$P_{\text{primitive}} = t\phi, \quad (1.10)$$

holds.

Therefore, introducing a timestep  $\tau$ , we can find the PDCM at all subsequent timesteps through the following convolution

$$\begin{aligned} \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; n\tau) = & \int \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_{1,1}, \mathbf{r}_{2,1}; \tau) \rho(\mathbf{r}_{1,1}, \mathbf{r}_{2,1} | \mathbf{r}_{1,2}, \mathbf{r}_{2,2}; \tau) \dots \rho(\mathbf{r}_{1,n-1}, \mathbf{r}_{2,n-1} | \mathbf{r}'_1, \mathbf{r}'_2; \tau) \times \\ & \prod_{k=1}^{n-1} \sqrt{g_{1,k} g_{2,k}} d\mathbf{r}_{1,k} d\mathbf{r}_{2,k}, \end{aligned} \quad (1.11)$$

where we denote with  $\mathbf{r}_{i,k}$  the position of particle  $i$  at timeslice  $k$ ,  $g_{i,k}$  is the determinant  $g$  calculated at  $\mathbf{r}_{i,k}$ ,  $d\mathbf{r} = d\theta d\varphi$ , and we choose  $\tau$  small enough so that

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau) \approx \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; \tau) e^{-\tau[\phi(\mathbf{r}_1, \mathbf{r}_2) + \phi(\mathbf{r}'_1, \mathbf{r}'_2)]/2}, \quad (1.12)$$

is a good approximation. We will perform the  $4(n-1)$  integrations in Eq. (1.11) through a simple quadrature method.

Even if in general we will be interested in the full  $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}'_1, \mathbf{r}'_2; t)$  in practice in a many body simulation we may use the *end point approximation* for the inter-action [6] where one only needs the diagonal components  $P(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = P(d(\mathbf{r}_1, \mathbf{r}_2); t)$ . Moreover on a sphere the distance between any two points  $\mathbf{r}_1$  and  $\mathbf{r}_2$  is bounded from above: the geodesic distance must be smaller than  $\pi a$  and the Euclidean distance than  $2a$ .

On the diagonal we will have

$$\rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t) = \rho(d(\mathbf{r}_1, \mathbf{r}_2); t) = \rho_{10}(\mathbf{0}, \mathbf{0}; t) \rho_{20}(\mathbf{0}, \mathbf{0}; t) e^{-P(d(\mathbf{r}_1, \mathbf{r}_2); t)}, \quad (1.13)$$

and  $\ln[2\pi t \sqrt{g(\mathbf{0})} \rho_{i0}(\mathbf{0}, \mathbf{0}; t)]$  is given up to orders  $t^8$  in Ref. [5] (see their Eq. (3.23)). For completeness we report their perturbative expansion in Eq. (A1) of Appendix A.

We will then tabulate

$$P(m\delta; n\tau) = -\ln \left[ \frac{\rho(m\delta; n\tau)}{\rho_0(m\delta; n\tau)} \right], \quad n = 1, 2, 3, \dots, \quad m = 1, 2, 3, \dots, m_{\text{max}} \quad (1.14)$$

for given small  $\tau$  and  $\delta$ . For an imaginary time  $t_n = n\tau$  this will require a number of  $4(n-1)m_{\text{max}}$  quadratures to calculate  $\rho(m\delta; n\tau) = \rho(d(\mathbf{r}_1, \mathbf{r}_2); t_n) = \rho(\mathbf{r}_1, \mathbf{r}_2 | \mathbf{r}_1, \mathbf{r}_2; t_n)$  for  $m = 1, 2, 3, \dots, m_{\text{max}}$ , through the convolution (1.11). Each convolution requires 2 quadratures over the azimuthal angle  $\varphi \in [-\pi, \pi[$  and the polar angle  $\theta \in [-\pi/2, \pi/2]$ . Choosing an angular discretization  $\delta_\varphi = 2\delta_\theta$  we will have that each angular integral reduces to a sum over  $l = \pi/\delta_\theta$  terms. Then in order to tabulate  $P(d; t_n)$  we need  $4(n-1)m_{\text{max}}l$  functions evaluations. This becomes more computationally demanding at later imaginary times  $t$  but in a many body simulation the pair-product inter-action is needed at a fixed small  $t$ , which determines the discretization error [6]. So one needs to tabulate it just once, at the beginning of the simulation.

In Table I we show  $P(m\delta; n\tau)$  for  $a = 1, m_1 = m_2 = m_e, Z_1 = -Z_2 = -1, \tau = 0.001, \delta = 0.1$  and  $n = 10$ .

TABLE I. Table for  $P(m\delta; n\tau)$  for  $a = 1, m_1 = m_2 = m_e, Z_1 = -Z_2 = -1, \tau = 0.001, \delta = 0.1$  and  $n = 10$ . For the quadratures we used Simpson rule with  $\delta_\theta = 0.01$  so that  $l = 314$ .

| $P_{\text{on}}^g$ | $P_{\text{on}}^e$ | $P_{\text{in}}$ | $m$ |
|-------------------|-------------------|-----------------|-----|
|                   |                   |                 | 1   |
|                   |                   |                 | 2   |
|                   |                   |                 | 20  |
|                   |                   |                 | 31  |

## II. THE CODE

We give here the FORTRAN code listing to generate the table of Eq. (1.14):

### Appendix A: $\rho_{i0}(\mathbf{0}, \mathbf{0}; t)$

Calling  $\mu_i = m_i/m_e$  and  $R = 2/a^2$  the Sphere scalar curvature, we find from equation (3.23) of Ref. [5]

$$\ln \left[ 2\pi t \sqrt{g(\mathbf{0})} \rho_{i,0}(\mathbf{0}, \mathbf{0}; t) \right] = \frac{tR_i}{12} + \frac{(tR_i)^2}{6!} \frac{1}{2} + \frac{(tR_i)^3}{9!} 16 + \frac{(tR_i)^4}{10!} \frac{59}{4} + \frac{(tR_i)^5}{11!} \frac{58}{3} + \frac{(tR_i)^6}{13!} \frac{550789}{1260} + \frac{(tR_i)^7}{14!} \frac{46838}{45} + \frac{(tR_i)^8}{17!} \frac{596004707}{720} + \mathcal{O}(t^9), \quad (\text{A1})$$

where  $R_i = R/\mu_i$ .

## AUTHOR DECLARATIONS

### Conflicts of interest

None declared.

### Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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None declared.

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